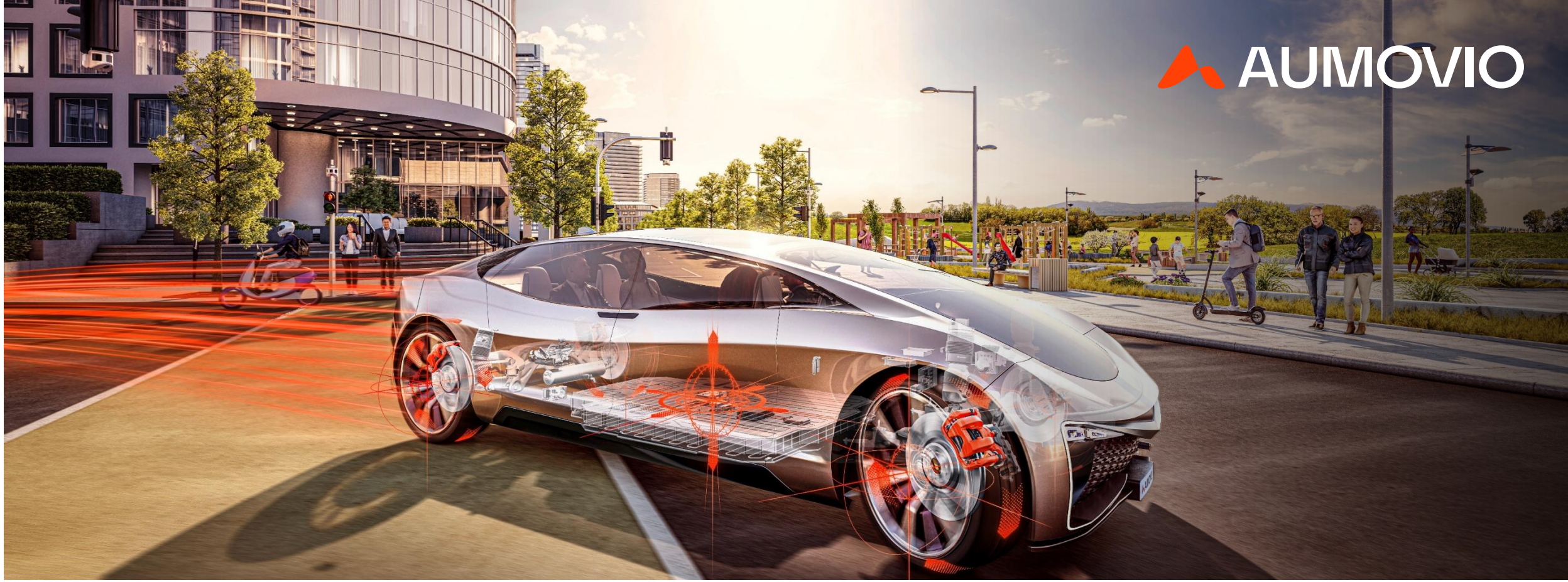


Competitor Finance Analysis

芜湖伯特利汽车安全系统股份有限公司

Company Code：603596

A SAM ASC BC Controlling, Feng Jingyi

2026/05/27

Safety & Motion

BTL Highlights



BTL Highlights

Key Figures

Sales Growth

CAGR: **36.2%**

2021-2025

Profitability

13.1%

Profit in 2025

Customer Concentration

73.4% Sales

from Top 5 Customers

New Projects

122 WCBS (one-box) Projects

New nominated in 2025

Production Capability

1.7m+1.1m

WCBS
Production Capability in 2026

Mar 2026 | EMB SOP

1st Global EMB

on Li Auto L9 Livis

Highlights

- **Sales** increased from 3.5 billion to 12 billion from 2021 to 2025, with **CAGR of 36.2%**
- **EBIT** remains stable at **over 13%** for the past 5 years
- **73.4%** of the sales from Top 5 customers. 2025 new customers include **Renault, Ford Europe, FVW and SVW, etc.**
- **WCBS Capacity**: 1.7 m current capacity +1.1 m additional Capacity in 2026)
- **EMB Capacity**: 600K in 2025, additional 1,200K in 2026. 10K SOP achieved in May 2026.

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Internal

One Page Y2025

BTL

	m CNY 2021	m CNY 2022	m CNY 2023	m CNY 2024	m CNY 2025	m CNY 2026 Q1	m CNY 2025 Q1
P&L							
Sales	**3,492**	**5,539**	**7,474**	**9,937**	**12,014**	**2,674**	**2,638**
Sales YoY		58.6%	34.9%	33.0%	20.9%		
GM	**845**	**1,243**	**1,635**	**2,101**	**2,360**	**520**	**474**
GM o.S%	24.2%	22.4%	21.9%	21.1%	19.6%	19.4%	18.0%
SG&A	**111**	**185**	**207**	**277**	**363**	**88**	**78**
SG&A o.S%	3.2%	3.3%	2.8%	2.8%	3.0%	3.3%	3.0%
R&D	**239**	**378**	**450**	**576**	**605**	**141**	**129**
R&D o.S%	6.9%	6.8%	6.0%	5.8%	5.0%	5.3%	4.9%
Profit	**610**	**784**	**1,046**	**1,373**	**1,568**	**325**	**307**
	17.5%	**14.2%**	**14.0%**	**13.8%**	**13.1%**	**12.2%**	**11.7%**

BS	2021	2022	2023	2024	2025	2026 Q1	2025 Q1
Inventory	425	899	1037	1420	1807	1849	1,567
	8.2	6.2	7.2	7.0	6.6		
AR	1854	2750	3947	5234	6017	5997	5,007
	53.1%	49.7%	52.8%	52.7%	50.1%		
AP	1348	2772	3637	4815	5166	4977	4,461
	38.6%	50.0%	48.7%	48.5%	43.0%		
Working Capital	932	877	1346	1839	2658	2870	2,114
	26.7%	15.8%	18.0%	18.5%	22.1%		
Debt to Assets Ratio	44.1%	47.2%	43.7%	45.7%	51.3%	50.0%	43.4%
Cash Flow before Financing	**-457**	**443**	**-163**	**352**	**-2889**	**-5**	**-121**

AR=AR/Sales
AP=AP/Sales
Inventory TR=Sales/Inventory

Liability /Asset= Total Liability / Total Asset
Working Capital= Inventory+ AR- AP
FCF=Net Operating cash flow- net investing cash flow

2025
2026

* Shareholding

2025-2026Q1 Key Investments & New Entities

198 mio (99%)*
Invest **Humanoid Robotics** Area
2025/07
Gongqingcheng Bojin Venture Capital

50 mio (1.22%)*
Invest **Low-Altitude Economy** Area
2025/11
Shanghai Volant Aerospace Technology

50 mio (2.78%)*
Invest **Humanoid Robotics** Area 2025/12
RealMan Robotics Company

60 mio(60%)*
Develop Ball Screws for Humanoid Robots Area
2025/09
Zhejiang Bojian Transmission Technology

35 mio (1.4%)*
Cooperate with Chery for Humanoid Robots Area
2025/12
AiMOGA Robot Company



1.12 billion (50.97%)*
Acquisition of **Yubei Steering**
2026/02
Complement steer-by-wire, bind global customers.

- Establish Wuhu Berthel Drive Technology Co., Ltd. (60mio CNY, 60% shareholding) → For **motor supply**
- Establish wholly owned subsidiary and new plant in Morocco for **North Afrika and Europe market** (75mio USD)
- Establish wholly owned subsidiary WBTL Germany GmbH in Frankfurt as the **European R&D Center**.(1mio USD)

2025 Project Status (2025YTD / 2026Q1)

Products	New SOP	New Awarded	In Development
EPB	125 / 6	218 / 56	201 /244
WCBS	53/ 9	122 /13	124 / 119
EMB			7 / 5

Comments

- **Sales:** 2025 sales rose 20.9% YoY to 12 billion CNY, driven by intelligent chassis product growth and overseas customer expansion.
- **Gross Margin:** Overall gross margin dipped 1.5pct YoY to 19.6%, pressured by price competition and lower overseas profitability.
- **R&D:** R&D investment reached 605mio with steady spending on steer-by-wire and EMB and humanoid robot core components.
- **Cash Flow:** Operating cash flow declined YoY amid heavy capital expenditure for capacity and global layout expansion.

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Internal

Company Introduction



Company Introduction

COMPANY DEVELOPMENT TIMELINE

2004 — Company Founded

2006 — Tangshan Bethel Established (Plant Launch)

2009 — BTL Electronics Founded

2013 — Weihai BTL Established (Plant Set-up)

BTL USA Established

2017 — Shanghai R&D Center

2018 — Listed on A-share Market（603596）

2021 — Mexico Lightweight Plant Construction Started

2022 — Acquired Wanda Steering Gear Co., Ltd.

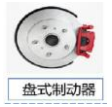
2024 — Wuhu BTL Automotive Suspension Technology Co., Ltd.

2025 — BTL Drive Technology Co., Ltd.

PRODUCT & TECHNOLOGY EVOLUTION

Technology Accumulation → **Lightweight & Electronic Braking** → **Brake-by-wire & Intelligence**

2005
Front disc brakes
Drum type brake



2006
Disc drum brake



2007-2008
Integrated Parking Brake
(first EPB sample)



2012-13
EPB/Vacuum booster

Aluminum steering knuckle
Hydraulic ESC





2019
Dual control EPB



2021
WCBS
(One-box)



2023
ADAS



2025
Air suspension



Future
PLG
EMB
…



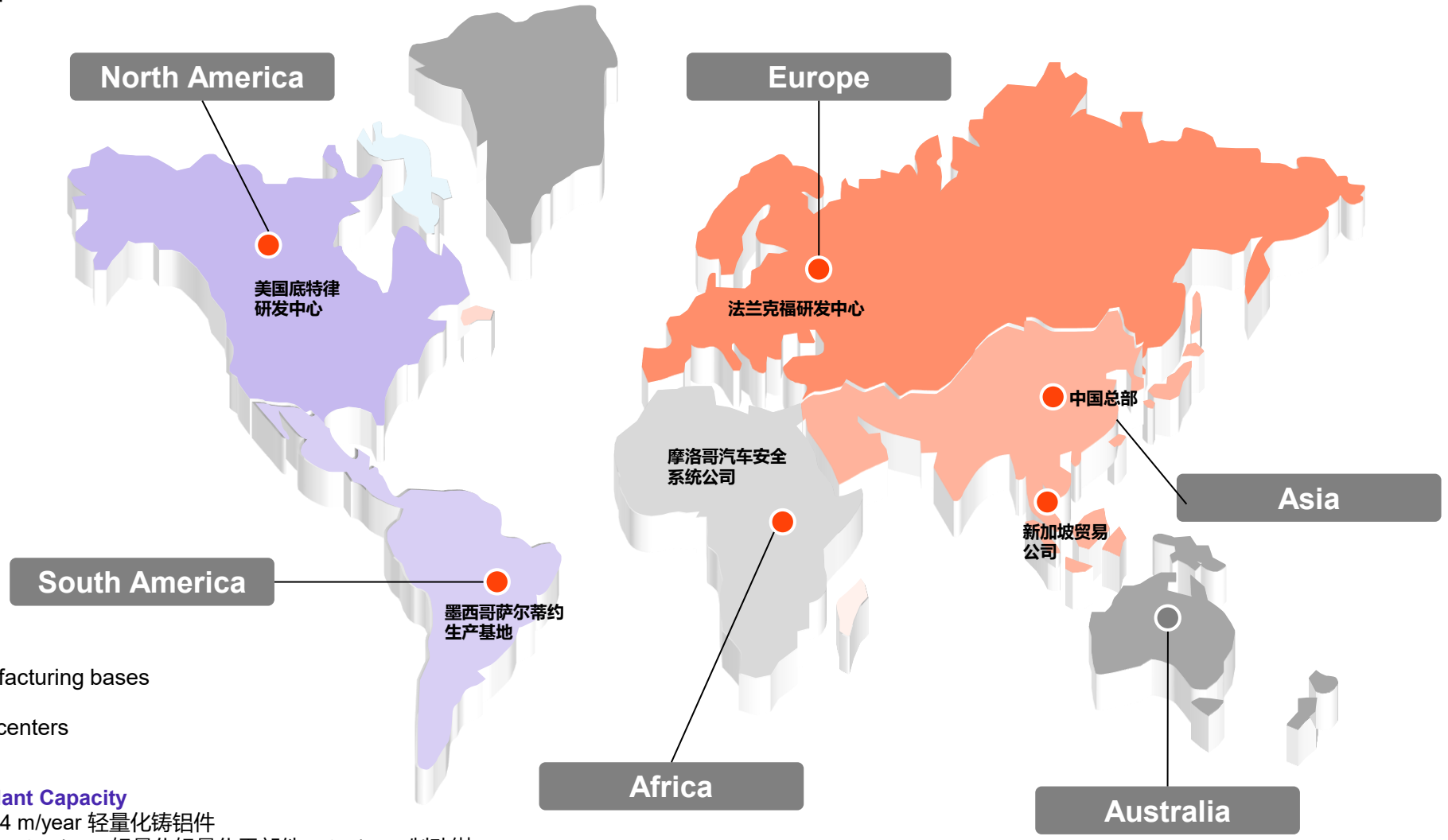
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Internal

Company Introduction

Footprint



18 Manufacturing bases

8 R&D centers

Mexico Plant Capacity
>Phase I: 4 m/year 轻量化铸铝件
>Phase II: 7.2 m/year 轻量化轻量化零部件 + 2m/year 制动钳

中国境内

安徽芜湖: *盘式制动器、真空助力器/主缸、ADAS、空气悬架、制动总装及研发中心*

山东威海: *铸铝转向节、控制臂、副车架等轻量化底盘件*

河北唐山:
盘式/鼓式制动器、驻车制动器、铝合金转向节

四川遂宁: *阀体机加、小总成*

(双利) 浙江台州: *智能底盘集成（与吉利合资，行泊一体方案）*

(万达) 浙江杭州: *机械转向器（MSG）、管柱、EPS 及线控转向*

上海: *研发技术中心*

中国境外

摩洛哥 丹吉尔/盖尼特拉:
汽车底盘与销售（服务于北美、欧洲客户）

墨西哥 圣路易斯波托西/萨尔蒂约:
铝合金轻量化零部件、制动系统

美国 底特律:
负责北美客户同步研发、测试及项目对接，为墨西哥工厂提供工程支持，无生产基地

德国: *法兰克福研发中心2025年底正式启用*

新加坡: *资本与贸易枢纽，本身不建工厂*

8个研发中心

中国: 芜湖 / 上海/ 杭州/ 苏州/ 西安/ 合肥
美国: 底特律
德国: 法兰克福

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Internal

Financial Part



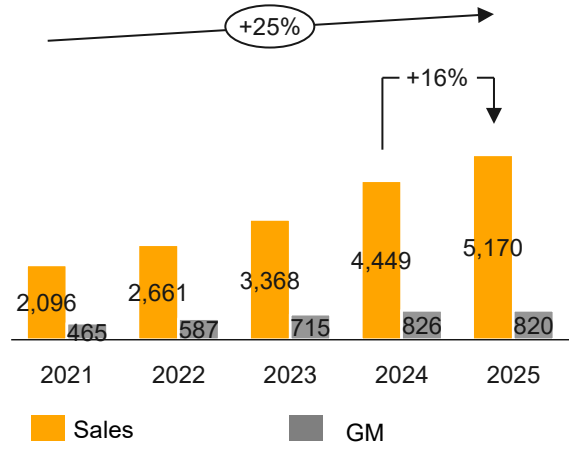
Financial Part

Sales & GM% by Products Groups



Mechanical Braking — Sales Share 2025

GM oS %	2021	2022	2023	2024	2025
	22.2%	22.0%	21.2%	18.6%	15.9%

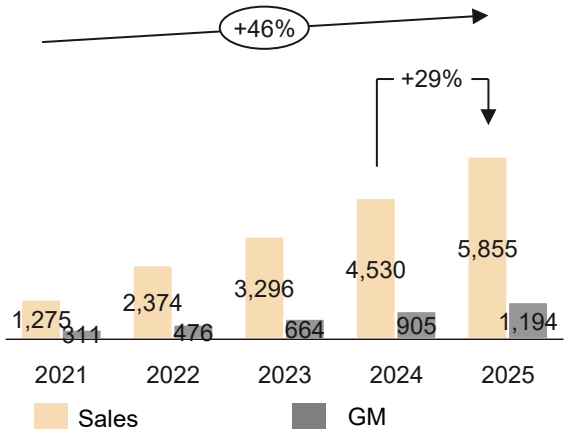


- Sales reached 5.17 billion, growing 16.21% YoY as a key revenue contributor.
- Gross margin fell to 15.87%, pressured by raw materials and market competition.



Intelligent Electronic-Control — Sales Share 2025

GM oS %	2021	2022	2023	2024	2025
	24.4%	20.0%	20.1%	20.0%	20.4%

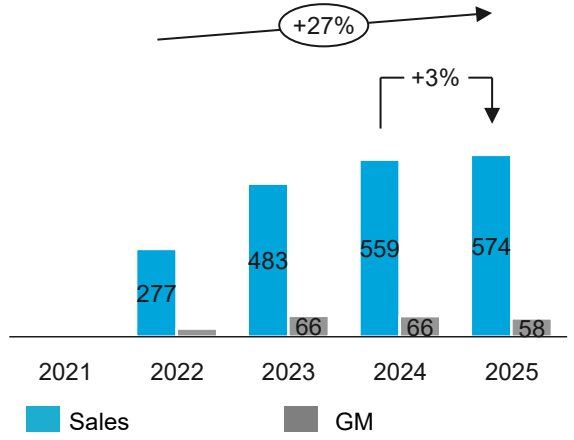


- Sales hit 5.855 billion (+29.26% YoY), ranking as the largest revenue segment.
- Gross margin remained stable at 20.40%, acting as core high-margin growth driver.



Mechanical Steering — Sales Share 2025

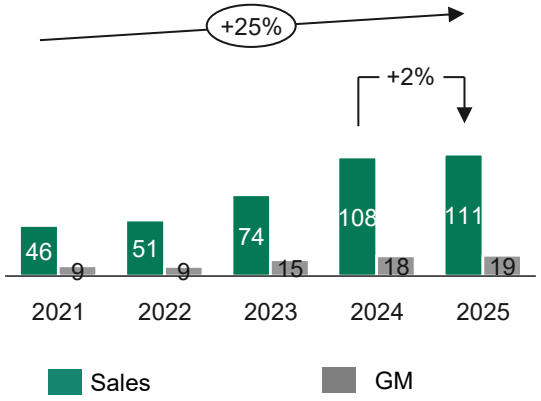
GM oS %	2021	2022	2023	2024	2025
	9.5%	13.7%	11.7%	10.2%	



- Sales rose slightly by 2.59% YoY to 574 mio with limited market demand.
- Gross margin dropped to 10.18% due to shrinking ICE vehicle demand and fierce price competition.

Others — Sales Share 2025

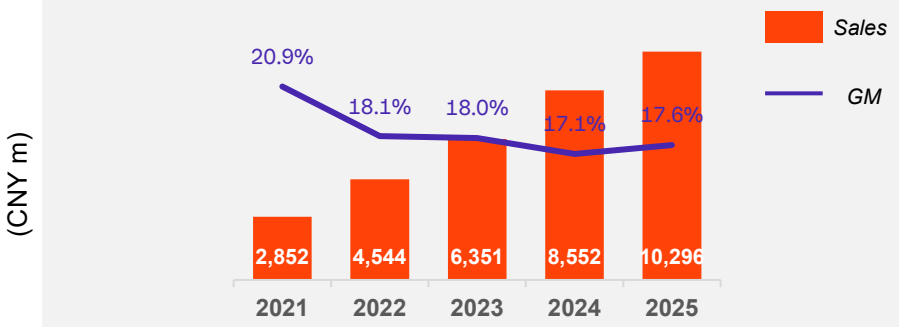
GM oS %	2021	2022	2023	2024	2025
	20.3%	17.2%	20.2%	16.9%	17.2%



Financial Part
Sales & GM% by Domestic & Aboard

Domestic

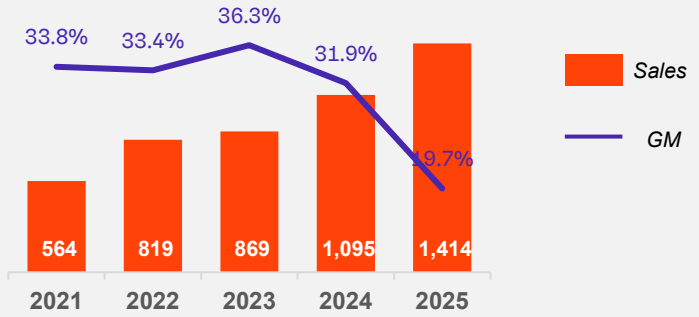
Sales Share 2025



- Sales accounted for **85.7%** of total sales in 2025, with steady gross margin of 17.61%, acting as core profit contributor.

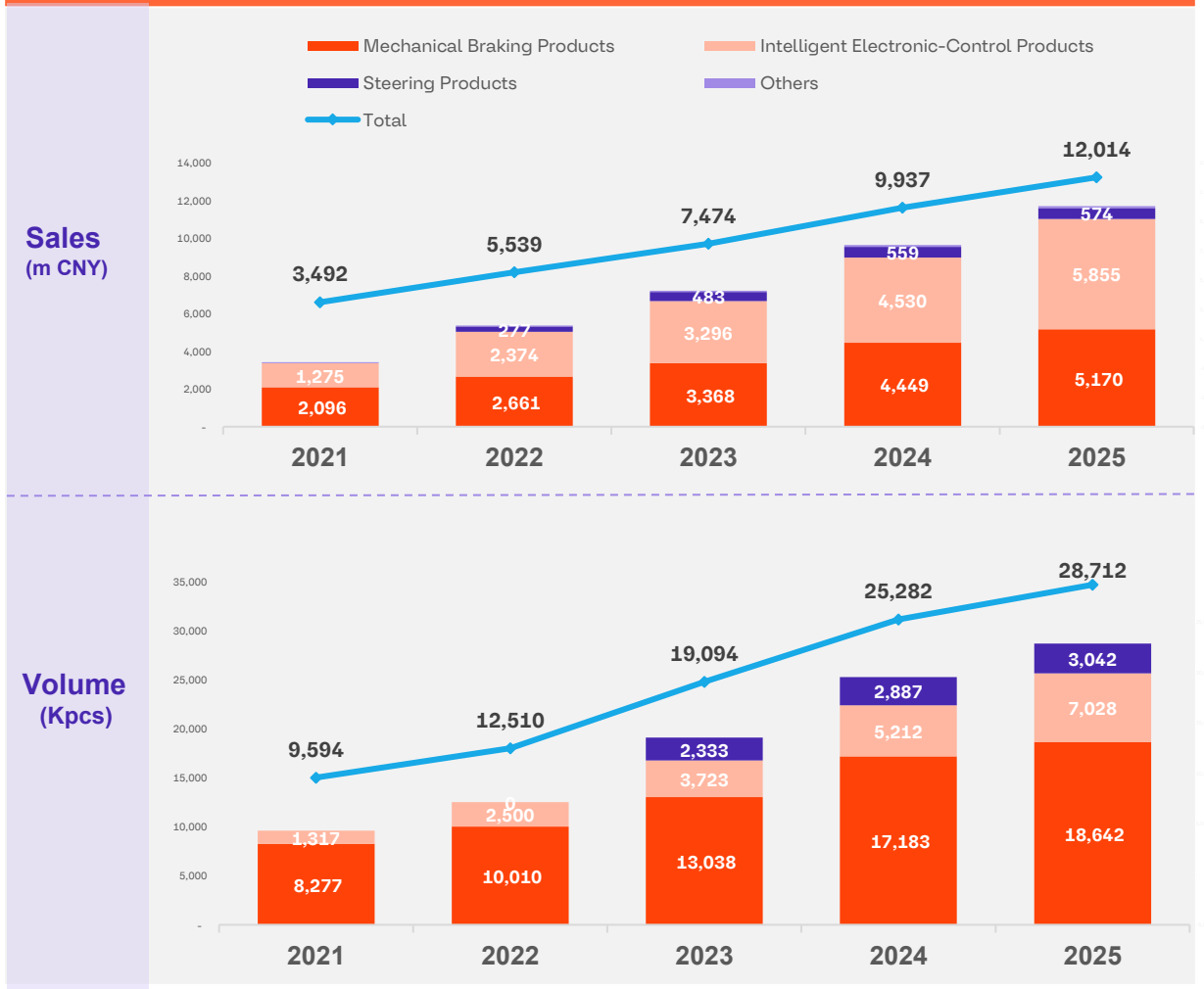
Oversea

Sales Share 2025



- Sales grew 29.17% YoY, while gross margin plunged from **31.9% (2024)** to **19.75% (2025)** due to capacity ramp-up costs and intensified competition.
- Profit of Mexico plant narrowed from -42.71% (2024) to **-7.83%** (2025). The plant remains in ramp-up, with profitability set to improve in 2026 via higher volume and cost optimization.

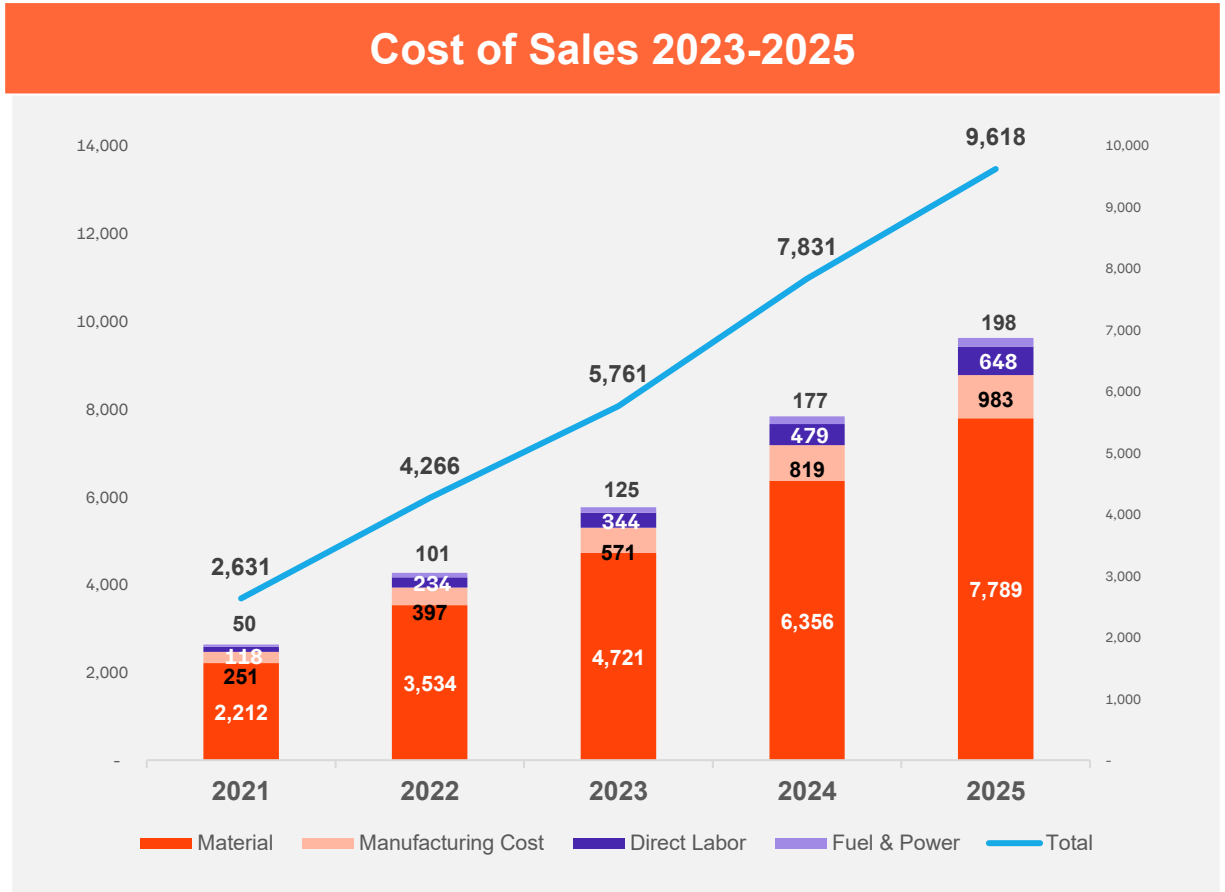
Total Sales by Product Groups



Internal

Financial Analysis
Cost Structure

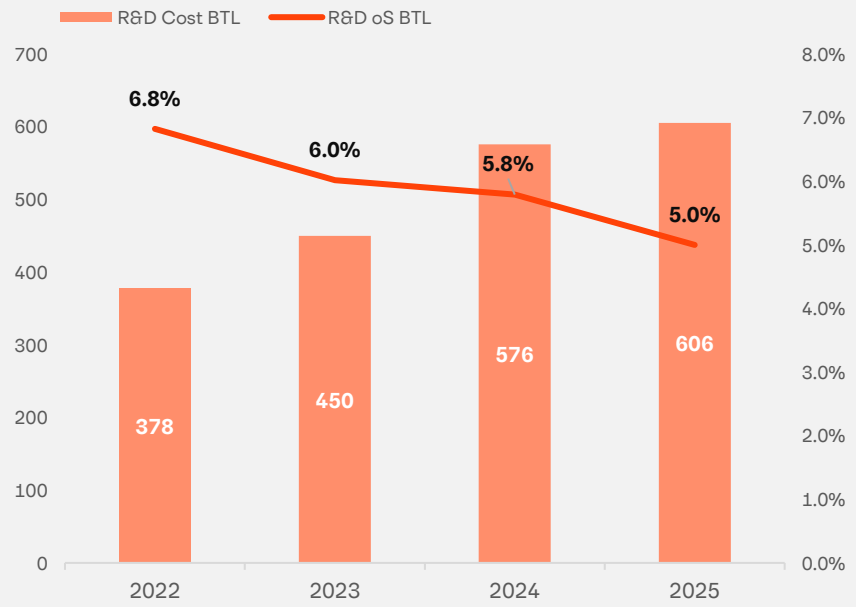
	m CNY 2021	m CNY 2022	m CNY 2023	m CNY 2024	m CNY 2025
Sales	**3,492**	**5,539**	**7,474**	**9,937**	**12,014**
Cost of Sales	**2,648**	**4,296**	**5,838**	**7,836**	**9,654**
Material	**2,212**	**3,534**	**4,721**	**6,356**	**7,789**
	63%	64%	63%	64%	65%
Manufacturing Cost	**251**	**397**	**571**	**819**	**983**
	7%	7%	8%	8%	8%
Direct Labor	**118**	**234**	**344**	**479**	**648**
	3%	4%	5%	5%	5%
Fuel & Power	**50**	**101**	**125**	**177**	**198**
	1%	2%	2%	2%	2%
Gross Margin	**845**	**1,243**	**1,635**	**2,101**	**2,360**



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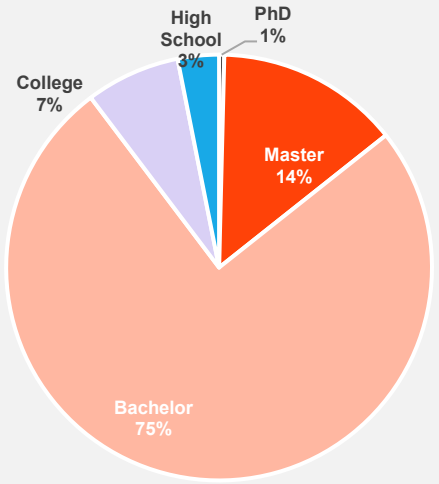
Financial Part
R&D Analysis

R&D Cost

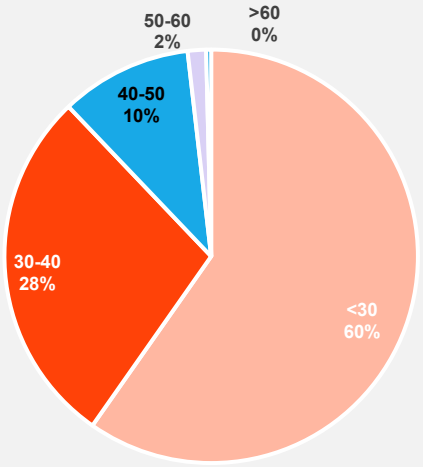


HC Status

Y2025 Education Background



Y2025 Age Structure



- Investment focused on core technologies including EMB, steer-by-wire and humanoid robot components。
- R&D oS is stable at around 6%,
- Strengthened core motor R&D via the 60%-held JV with Jinrun Electric.

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Internal

HC

R&D Headcount	2021	2022	2023	2024	2025	2025 vs 2024
	481	854	1,083	1,250	1,535	285
		78%	*27%*	*15%*	*23%*	
Education Background	**2021**	**2022**	**2023**	**2024**	**2025**	
PhD	6	6	9	7	6	(1)
Master	76	113	144	170	214	44
Bachelor	281	552	734	953	1,157	204
College	50	112	129	101	110	9
High School	68	71	67	19	48	29
Age Structure	**2021**	**2022**	**2023**	**2024**	**2025**	
<30	216	425	575	673	917	244
30-40	195	316	377	413	432	19
40-50	56	89	106	134	158	24
50-60	14	22	22	26	22	(4)
>60	-	2	3	4	6	2

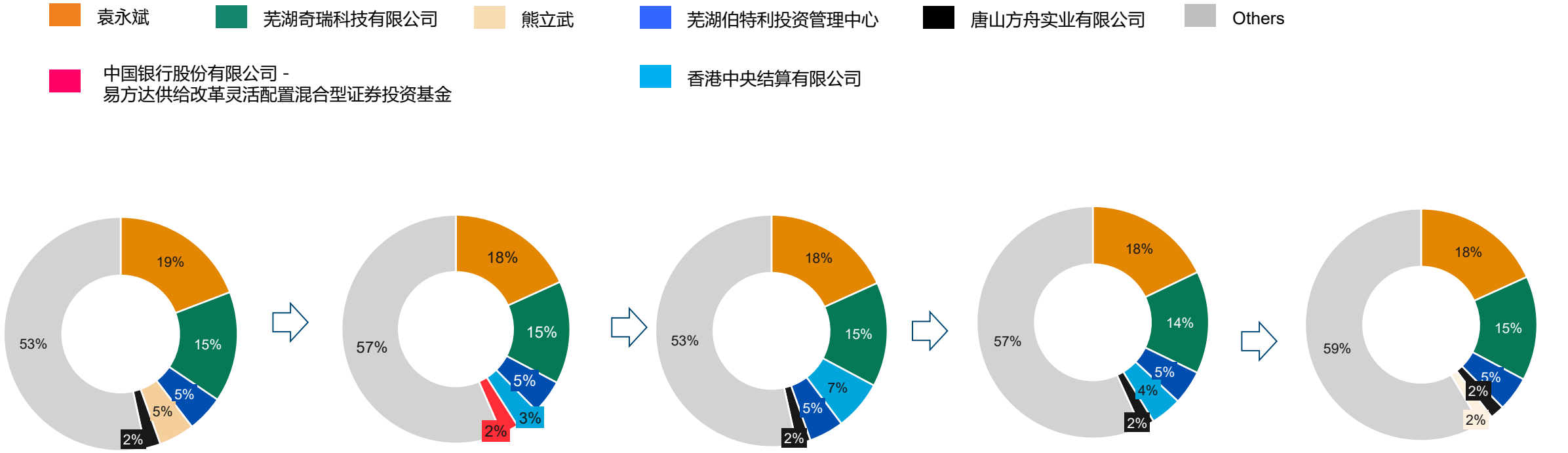
Total Headcount	2021	2022	2023	2024	2025	2025 vs 2024
Total HC	**1,912**	**3,992**	**4,560**	**5,932**	**6,358**	**426**
		109%	*14%*	*30%*	*7%*	
Department						
Production	1,071	2,443	2,873	3,540	3,750	210
Sales	30	50	44	52	57	5
R&D& Technical	541	1,145	1,278	1,885	2,140	255
Finance	35	56	58	73	82	9
Admin	235	298	307	382	329	-53
Education Background						
PhD	6	6	17	7	6	-1
Master	88	126	151	190	245	55
Bachelor	454	813	1,027	1,396	1,831	435
College	229	525	695	1,084	850	-234
High School	1,135	2,522	2,670	3255	3426	171

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Equity Structure Timeline

上市时间：2018.04.27



总股东数：15,448

向总经理颜士富先生(原**博世汽车部件苏州有限公司副总裁**)授予416,000 股限制性股票

总股东数：16,919

颜士富离开，创立**“坐标系智能科技”**，专注纯机电线控（EMB）系统，任 CEO

总股东数：20,883

海外投资者持股持续增多，占比 7%

总股东数：43,751

散户股东暴增，股东总数翻倍

总股东数：65,182

2026/02：单月增长1.84万散户
- 收购豫北转向50.97%股权
- 设立德国子公司

海外减持：剔除MSCI指数

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Internal

Supplementary Information (HK IPO Prospectus)

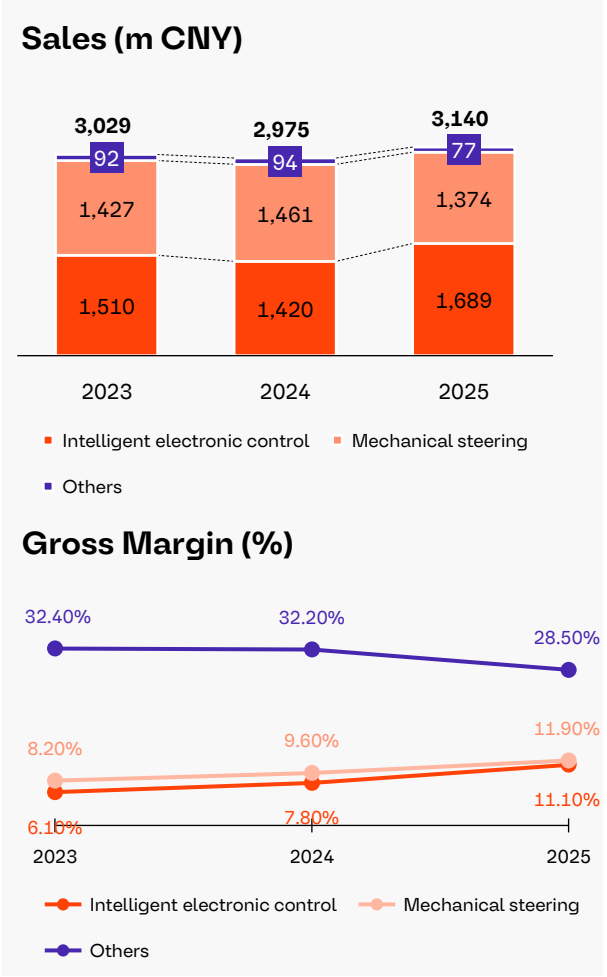
Supplementary Information (HK IPO Prospectus)

Yubei Steering

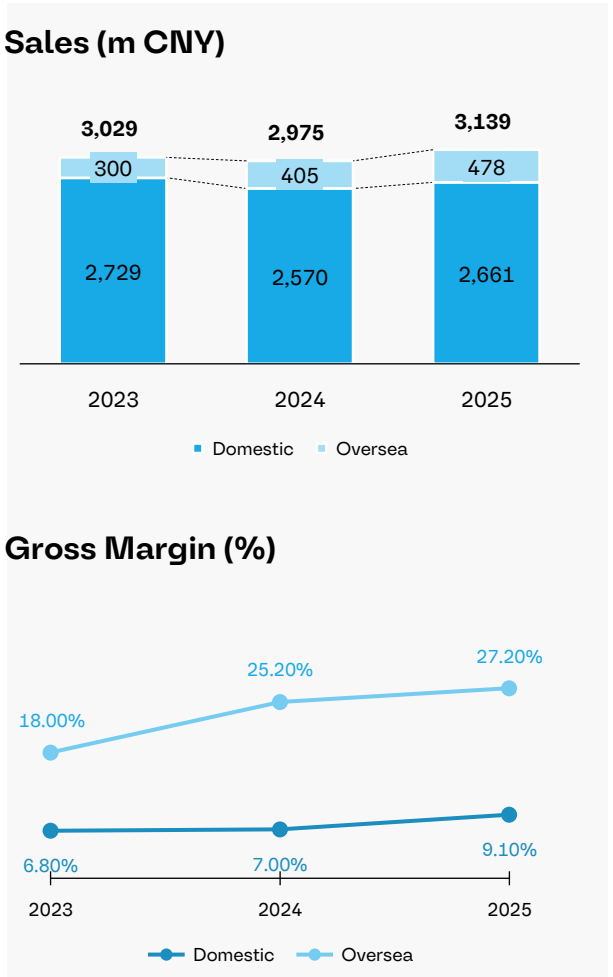
P&L (m CNY)	m CNY 2023	m CNY 2024	m CNY 2025
Sales	**3,029**	**2,975**	**3,139**
Sales YoY		*-1.8%*	*5.5%*
Material	2,265	2,248	2,251
o.S%	*74.8%*	*75.6%*	*71.7%*
Direct labor	210	213	214
o.S%	*6.9%*	*7.1%*	*6.8%*
Manufacturing costs	223	137	141
o.S%	*7.4%*	*4.6%*	*4.5%*
*Others**	93	94	160
o.S%	*3.1%*	*3.2%*	*5.1%*
Cost of Sales	**2,790**	**2,692**	**2,767**
Cost of Sales %	*92.1%*	*90.5%*	*88.1%*
Gross Margin	**238**	**283**	**372**
Gross Margin %	*7.9%*	*9.5%*	*11.9%*
there of R&D	*81*	*80*	*126*
o.S%	*2.7%*	*2.7%*	*4.0%*
there of SG&A	*165*	*173*	*194*
o.S%	*5.5%*	*5.8%*	*6.2%*
Net Profit	**90**	**134**	**157**
Net Profit %	*3.0%*	*4.5%*	*5.0%*

*Other mainly included logistics fees, warranty costs and energy and fuel costs.

Product Groups



Domestic & Oversea



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Internal

Supplementary Information (HK IPO Prospectus)

Capacity & Utilization Rate

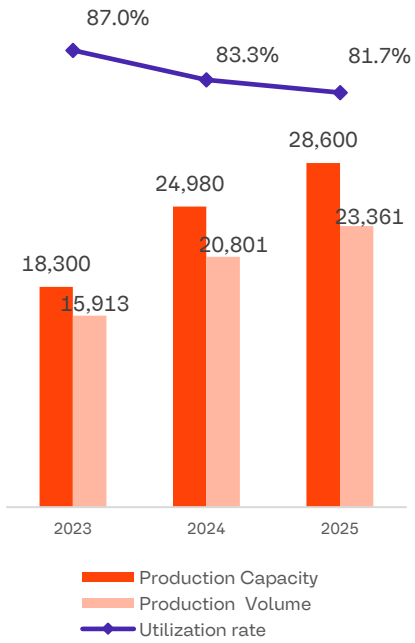
BTL Capacity (K Unit)



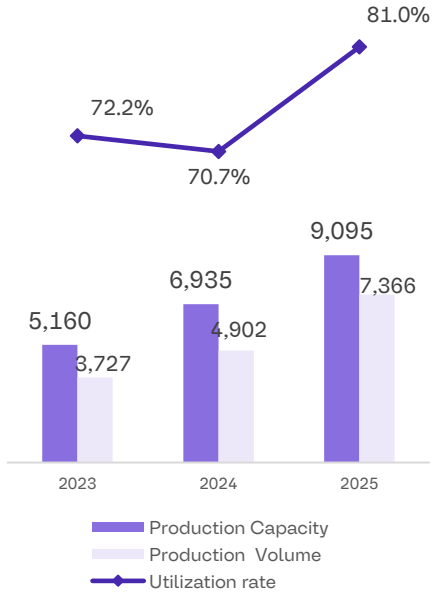




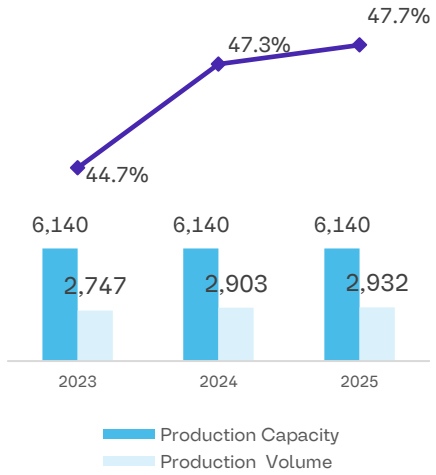
Mechanical Braking



Intelligent Electronic-Control



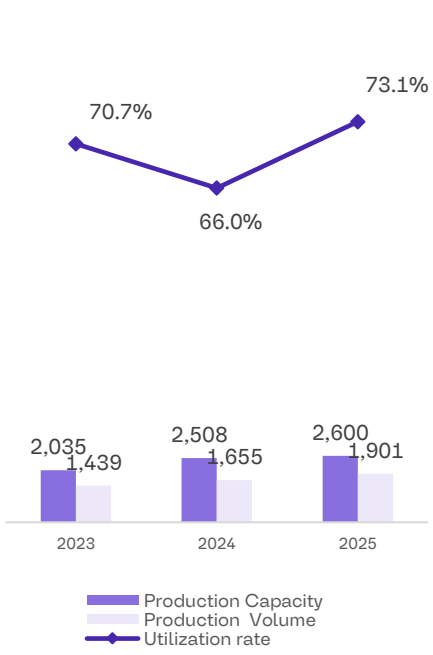
Mechanical Steering



Yubei Steering Capacity (K Unit)

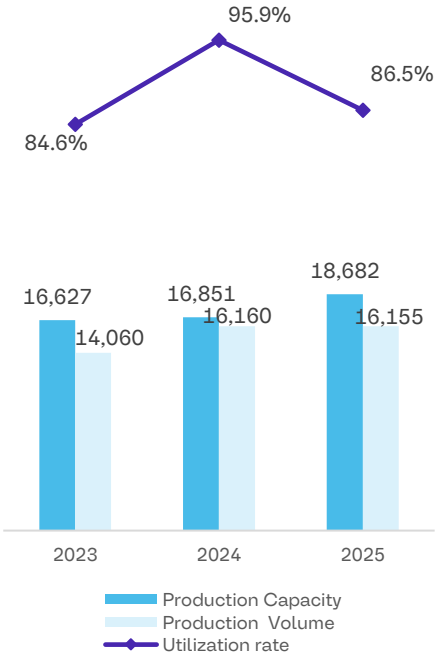


Intelligent Electronic-Control





Mechanical Steering



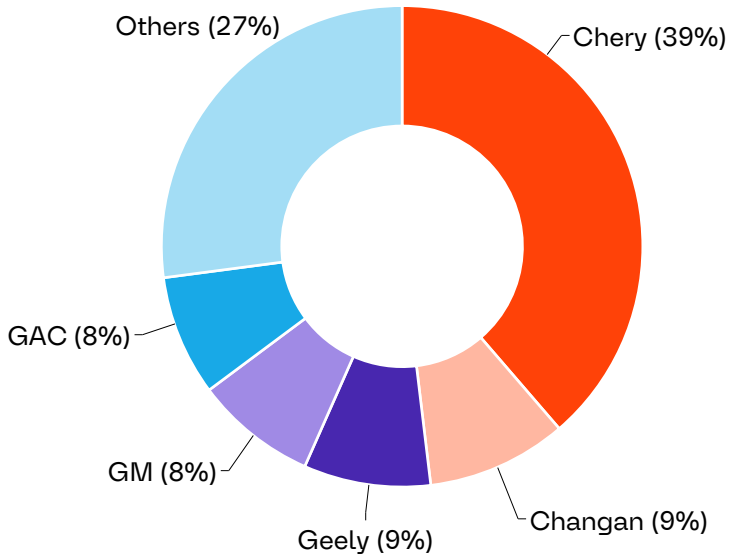
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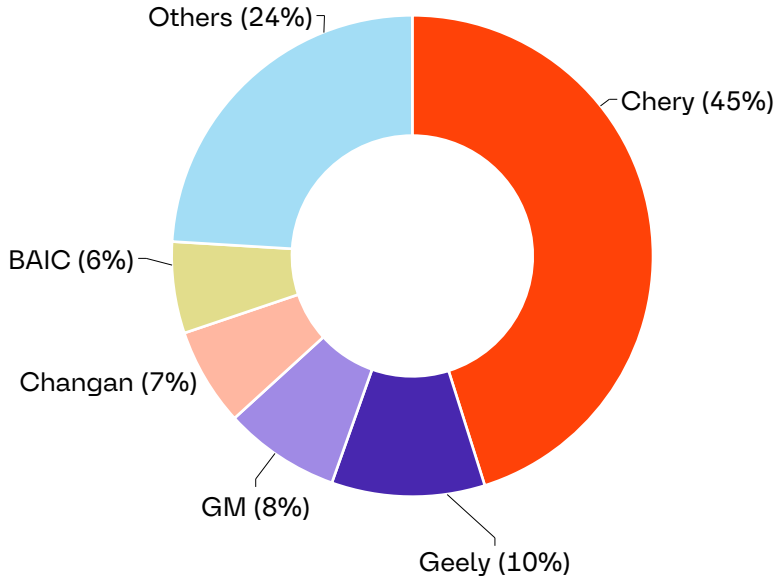
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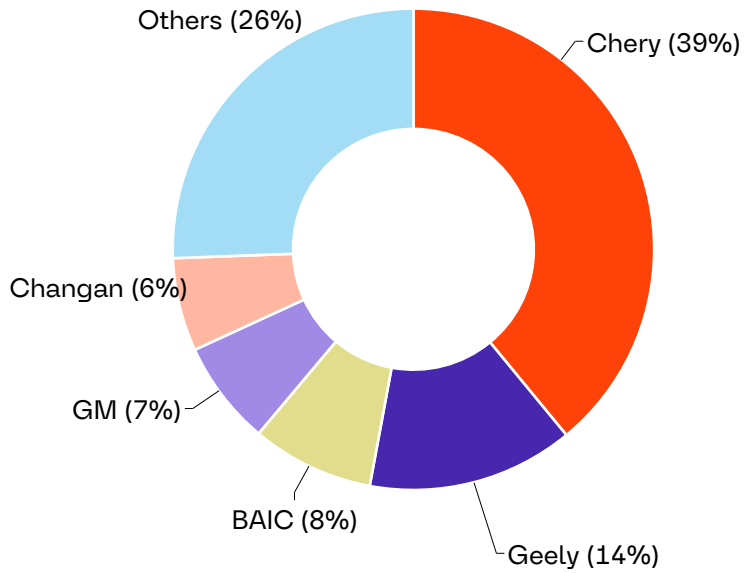
Supplementary Information (HK IPO Prospectus)

Sales Top 5 Customer Share%

	2023	2024	2025
Sales Top 5 Customer Share %	72.9%	76.0%	74.4%
Sales Top 1 Customer Share %	38.7%	45.1%	39.0%







2023 → **2024** → **2025**

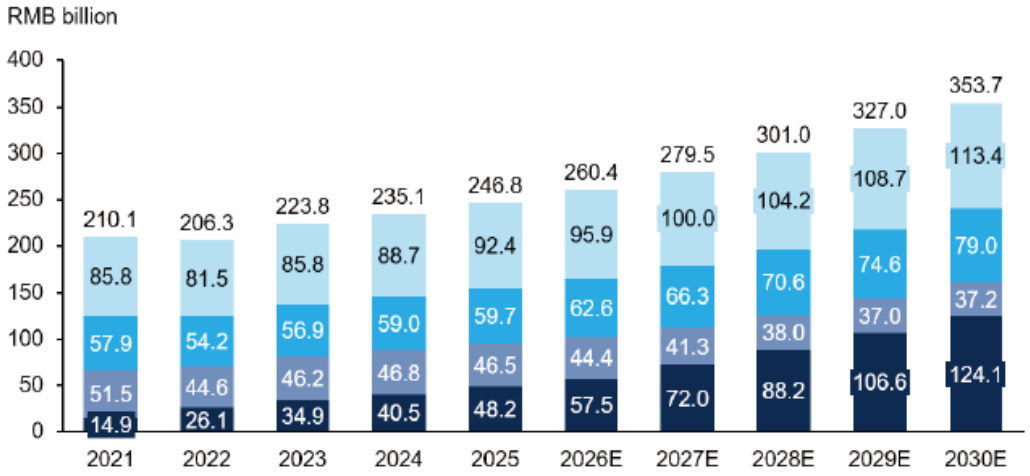
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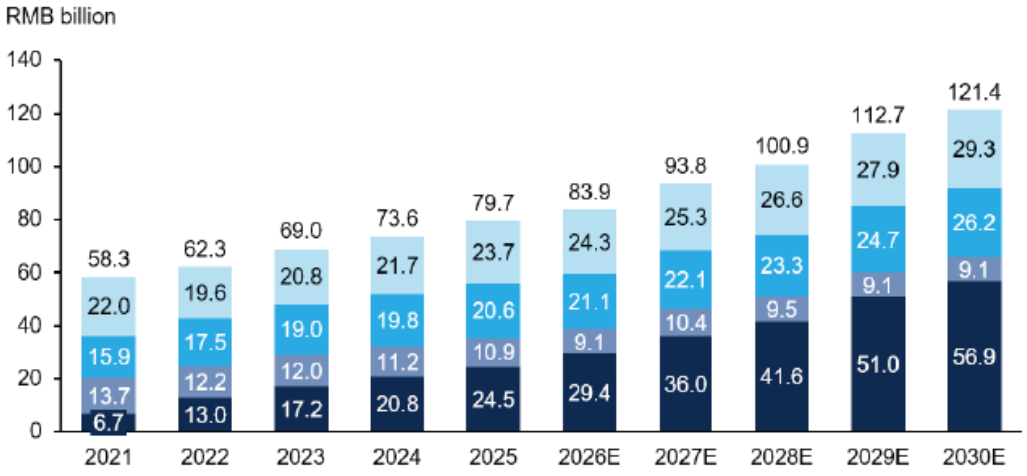
Supplementary Information (HK IPO Prospectus)

Market Size of Global and China's Automotive Braking System Industry by Revenue, 2021–2030E

	CAGR 2021-2025	CAGR 2026E-2030E
Brake	1.9%	4.3%
EPB	0.8%	6.0%
ABS、ESP	-2.5%	-4.3%
BbW	34.1%	21.2%
Global Automotive Braking Systems	**4.1%**	**8.0%**



	CAGR 2021-2025	CAGR 2026E-2030E
Brake	1.9%	4.7%
EPB	6.6%	5.5%
ABS、ESP	-5.7%	-0.1%
BbW	38.4%	18.0%
China's Automotive Braking Systems	**8.1%**	**9.7%**

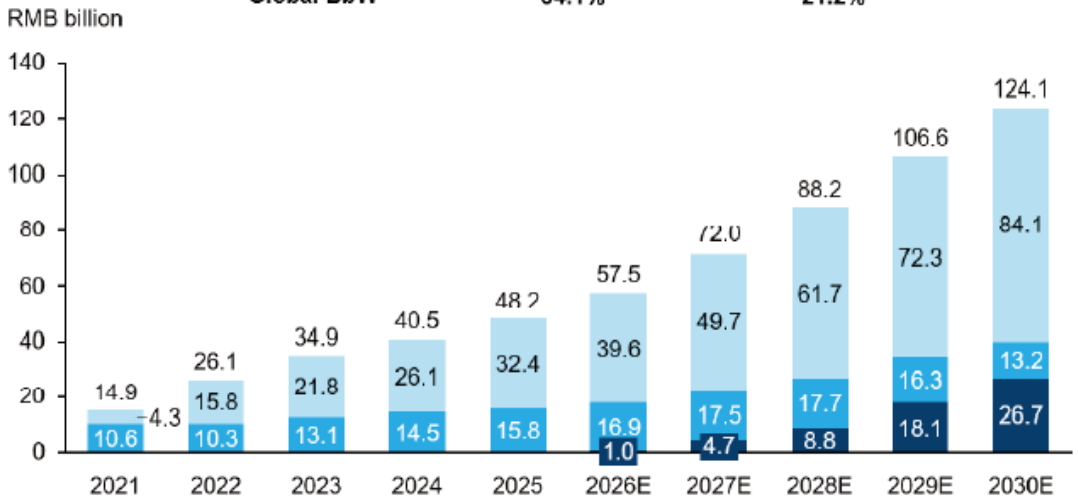


Source: CAAM, Frost & Sullivan

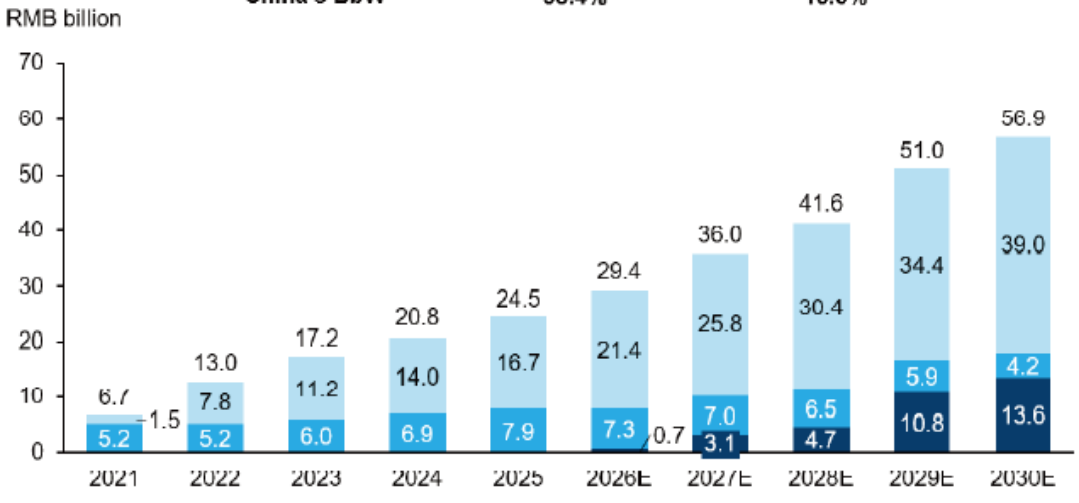
Supplementary Information (HK IPO Prospectus)

Market Size of Global and China's Automotive BbW Industry by Revenue, 2021–2030E

	CAGR 2021-2025	CAGR 2026E-2030E
One-box	65.7%	20.7%
Two-box	10.5%	-5.9%
EMB		125.6%
Global BbW	**34.1%**	**21.2%**



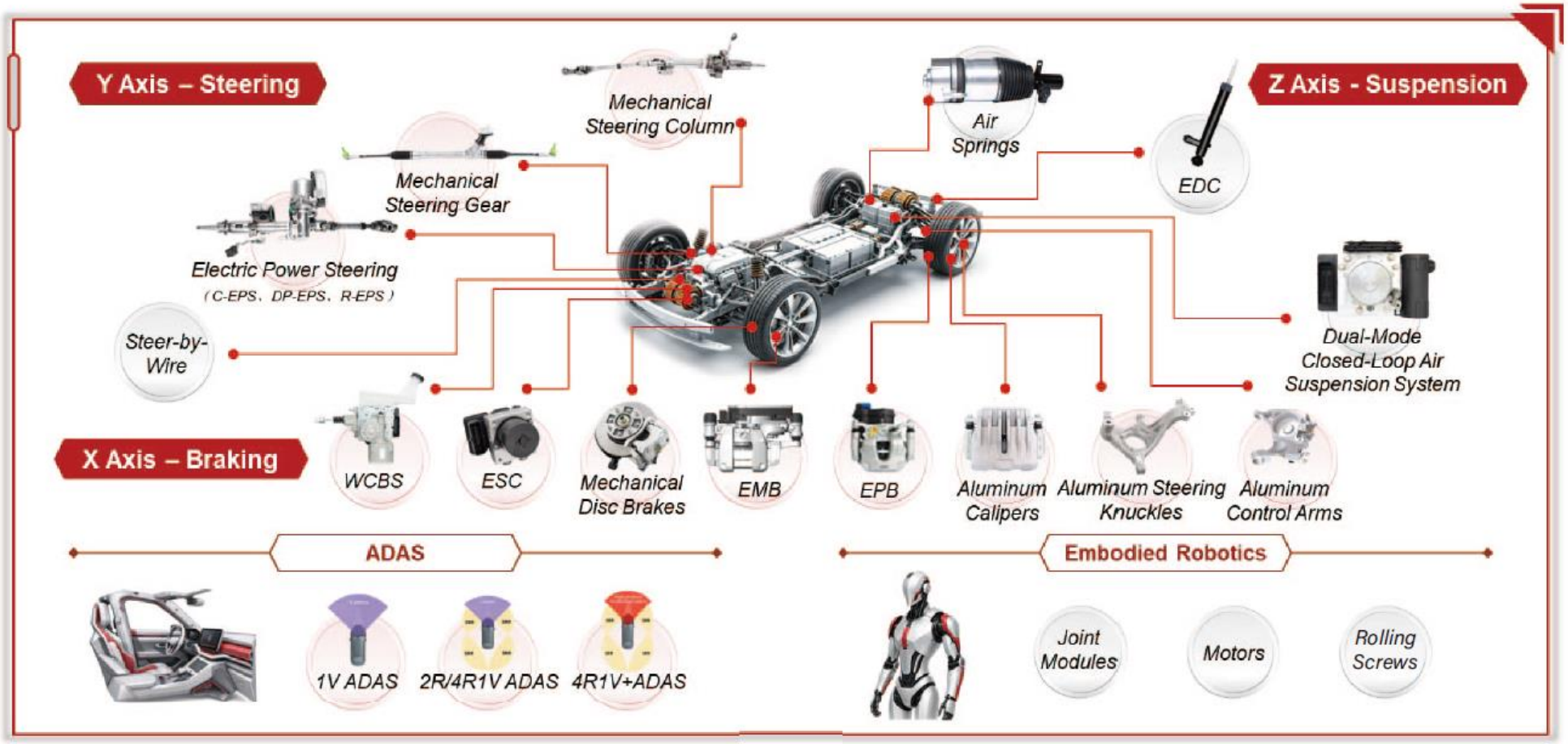
	CAGR 2021-2025	CAGR 2026E-2030E
One-box	81.8%	16.2%
Two-box	11.1%	-12.6%
EMB		111.1%
China's BbW	**38.4%**	**18.0%**



Source: CAAM, Frost & Sullivan

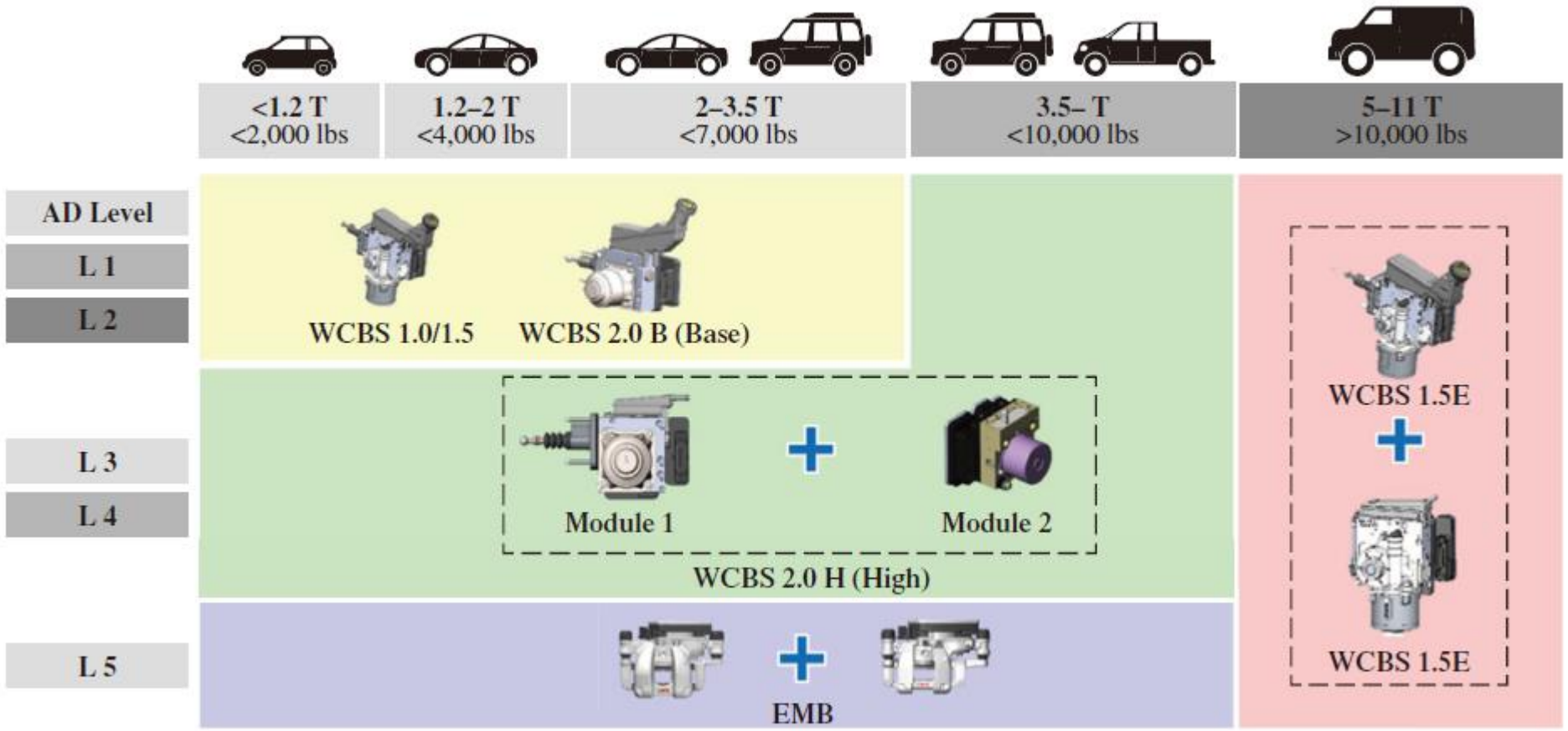
Supplementary Information (HK IPO Prospectus)

Product Portfolio



Supplementary Information (HK IPO Prospectus)

WCBS (One-box) Product Portfolio



BbW (Brake-by-Wire) primarily follows two technological pathways: EHB and EMB.

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Supplementary Information (HK IPO Prospectus)

Full List of Intermediaries – Hong Kong IPO

Submitted to HKEX on May 21, 2026

Sole Sponsor: CICC (中金)

Industry Consultant: Frost & Sullivan

Reporting Accountant: RSM (容诚会计师事务所)

PRC Legal Counsel (Issuer): Jingtian & Gongcheng

HK & US Legal Counsel (Issuer): Baker McKenzie

Sponsor's Legal Counsel

- PRC: Commerce & Finance Law Offices
- HK & US: Deacons

Compliance Advisor: Hongbo Capital

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Thank You!

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